

LECTURE 3: EVALUATION

METRICS THAT MATTER FOR MEDICAL IMAGING

Mehmet Kurt, Tianyi Ren
University of Washington

August 24, 2026

Evaluation, and what to expect

TERMINOLOGY

		Prediction	
		0	1
Ground truth	0	TN m_{00}	FP m_{01}
	1	FN m_{10}	TP m_{11}

↑ missed disease

Be careful: some packages transpose this layout.

- ▶ **True positive** (m_{11}): an example of class 1 predicted as class 1.
- ▶ **False positive** (m_{01}): class 0 predicted as class 1. **Type I error.**
- ▶ **True negative** (m_{00}): class 0 predicted as class 0.
- ▶ **False negative** (m_{10}): class 1 predicted as class 0. **Type II error.**

Total instances $m = m_{00} + m_{01} + m_{10} + m_{11}$, and the **error rate** is $(m_{01} + m_{10})/m$.

Error rate hides imbalance. If 10% of examples are class 1 and 90% are class 0, calling *everything* class 0 already scores a 10% error rate. On a disease with 5% prevalence, always predicting “healthy” reaches **95% accuracy** and catches **zero** patients: every error it makes is an m_{10} , the one cell that matters clinically.

COMMON MEASURES

What machine learning calls them

- ▶ Accuracy = $\frac{TP + TN}{TP + FP + FN + TN}$
- ▶ Precision = $\frac{TP}{TP + FP}$ of the declared positives
- ▶ Recall = $\frac{TP}{TP + FN}$ of the actual positives
- ▶ $F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

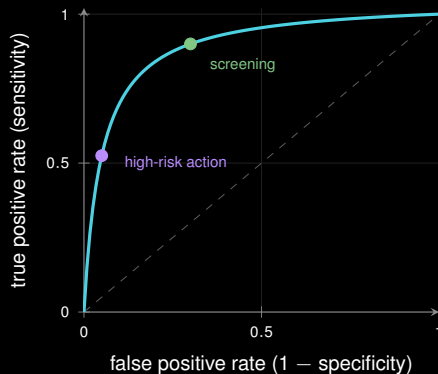
What medicine calls them

- ▶ Sensitivity = $\frac{TP}{TP + FN}$ the same as recall
- ▶ Specificity = $\frac{TN}{FP + TN}$ of the actual negatives
- ▶ False positive rate = $\frac{FP}{FP + TN}$
- ▶ PPV = precision, NPV = $\frac{TN}{TN + FN}$

Two literatures, one set of numbers

Sensitivity is recall. Text classification reports *precision* and *recall*; medicine reports *PPV* and *sensitivity*. And none of these is a fixed property of the model: each one moves with the **threshold** you pick for calling something positive.

ROC, AUC, AND CHOOSING WHERE TO SIT ON THE CURVE



The curve

Sweep the threshold from 1 to 0 and plot sensitivity against 1—specificity. **AUC** summarizes the whole curve in one number — threshold-free, good for comparing models.

The operating point

AUC does not deploy. You must pick a point: **high sensitivity** for screening (a miss is worse than a callback), **high specificity** before a biopsy or surgery.

Two footnotes

Youden's index ($J = \text{sens} + \text{spec} - 1$) picks a balanced point when you have no preference. At very low prevalence, prefer the **precision–recall** curve.

PITFALLS, AND WHAT TO EXPECT

Overfitting

Train accuracy $\gg$ validation accuracy.
Reach for augmentation, weight decay, early stopping, and pretrained weights — in that order.

Class imbalance

The rare class gets ignored and the loss looks fine. Use class weights or resampling, and always read *per-class* metrics.

Leakage

The same patient in train and test inflates every number you report. Split by patient. Check it explicitly — do not assume the dataset did it.

What to expect

On a well-posed task with a few thousand images per class, a fine-tuned ResNet-18 reaches accuracy in the high 80s to low 90s (%) within a few epochs — and a clear margin over the same network trained from scratch.